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(54) **SELECTING NETWORK ROUTES BASED  
ON AGGREGATING MODELS THAT  
PREDICT NODE ROUTING PERFORMANCE**

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See application file for complete search history.

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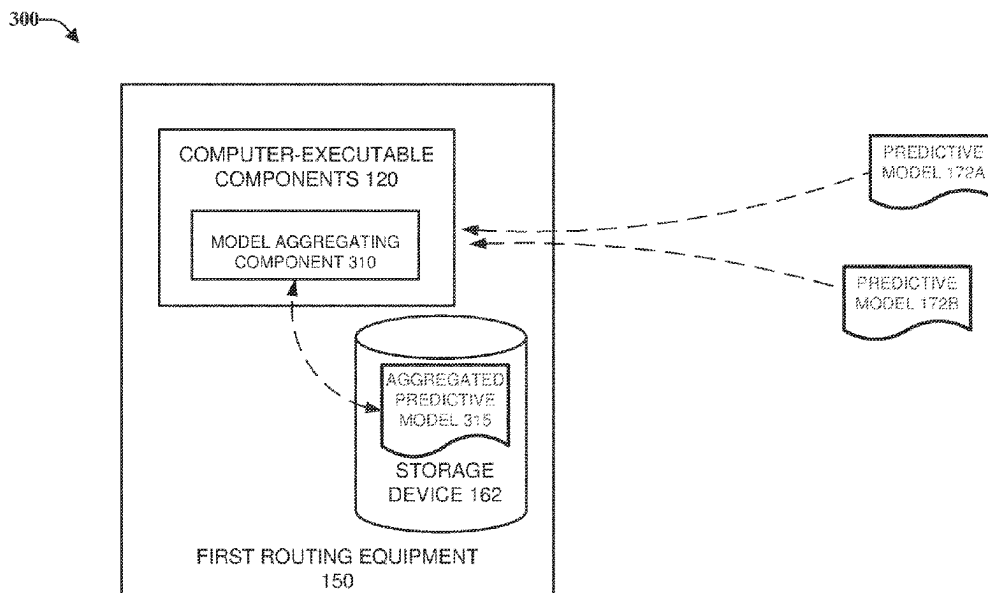
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Primary Examiner — Karen C Tang

(57) **ABSTRACT**

The technologies described herein are generally directed to selecting network routes based on aggregating models that can predict routing performance in a fifth generation (5G) network or other next generation networks. For example, a method described herein can include communicating, to second routing equipment, a first model describing a delay predicted to be caused to a future communication by the future communication being transited via the first routing equipment. The method can further include receiving, from the second routing equipment, a current communication for transit via the first routing equipment to destination equipment, wherein the first routing equipment was selected by the second routing equipment based on the first model, and second models, other than the first model, describing respective predicted delays from other routing equipment other than the first routing and second routing equipment.

**20 Claims, 10 Drawing Sheets**



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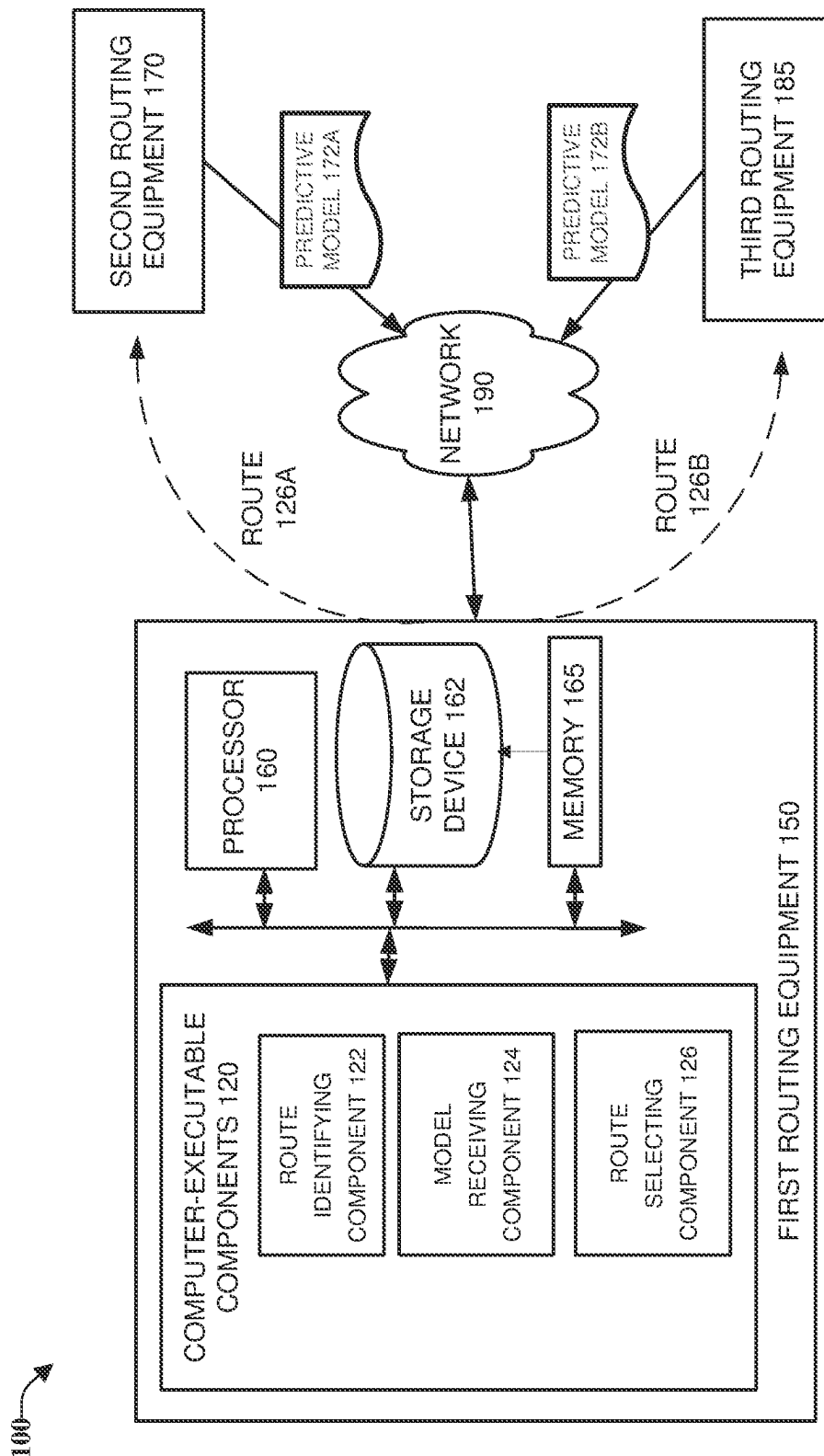


FIG. 1

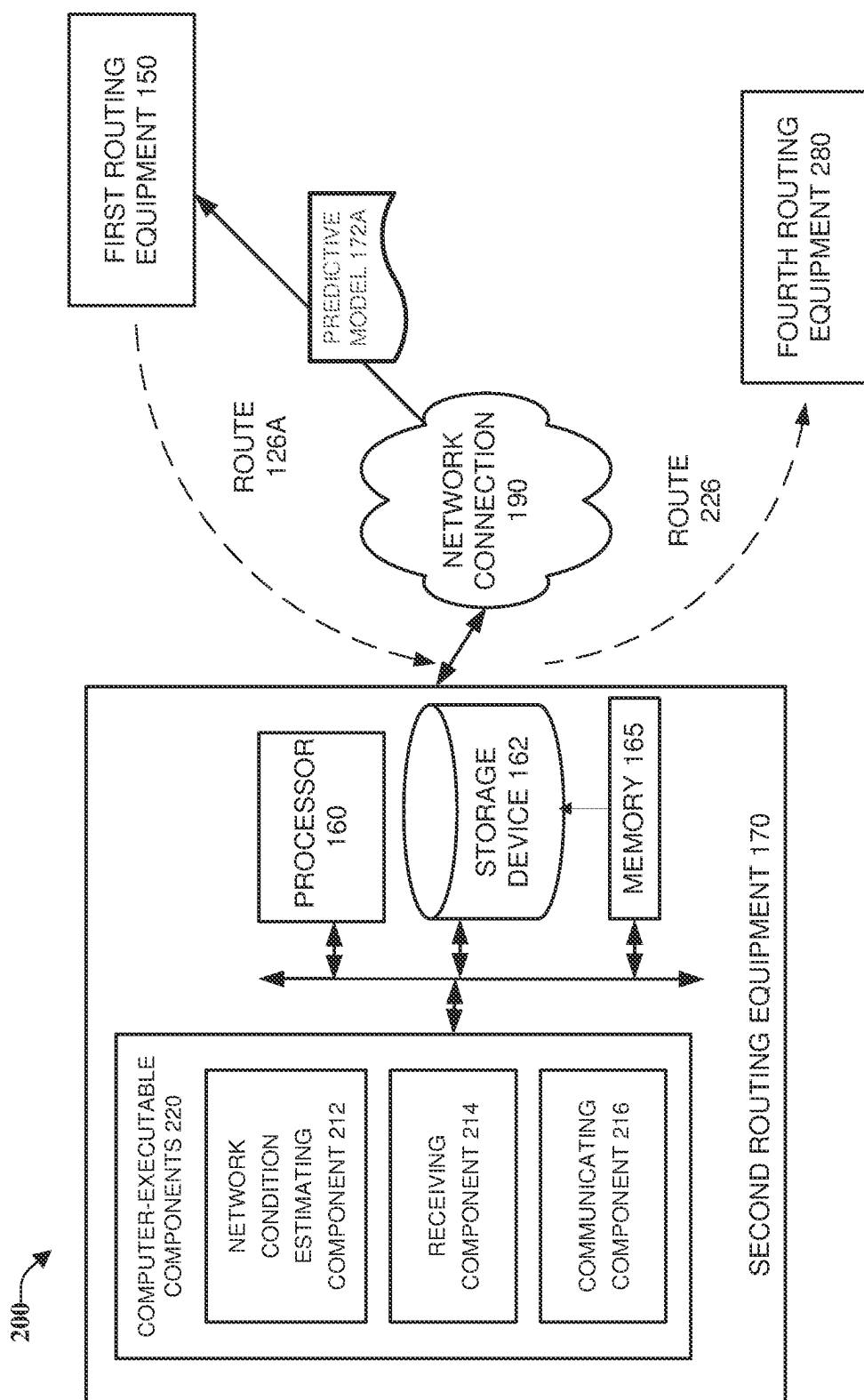


FIG. 2

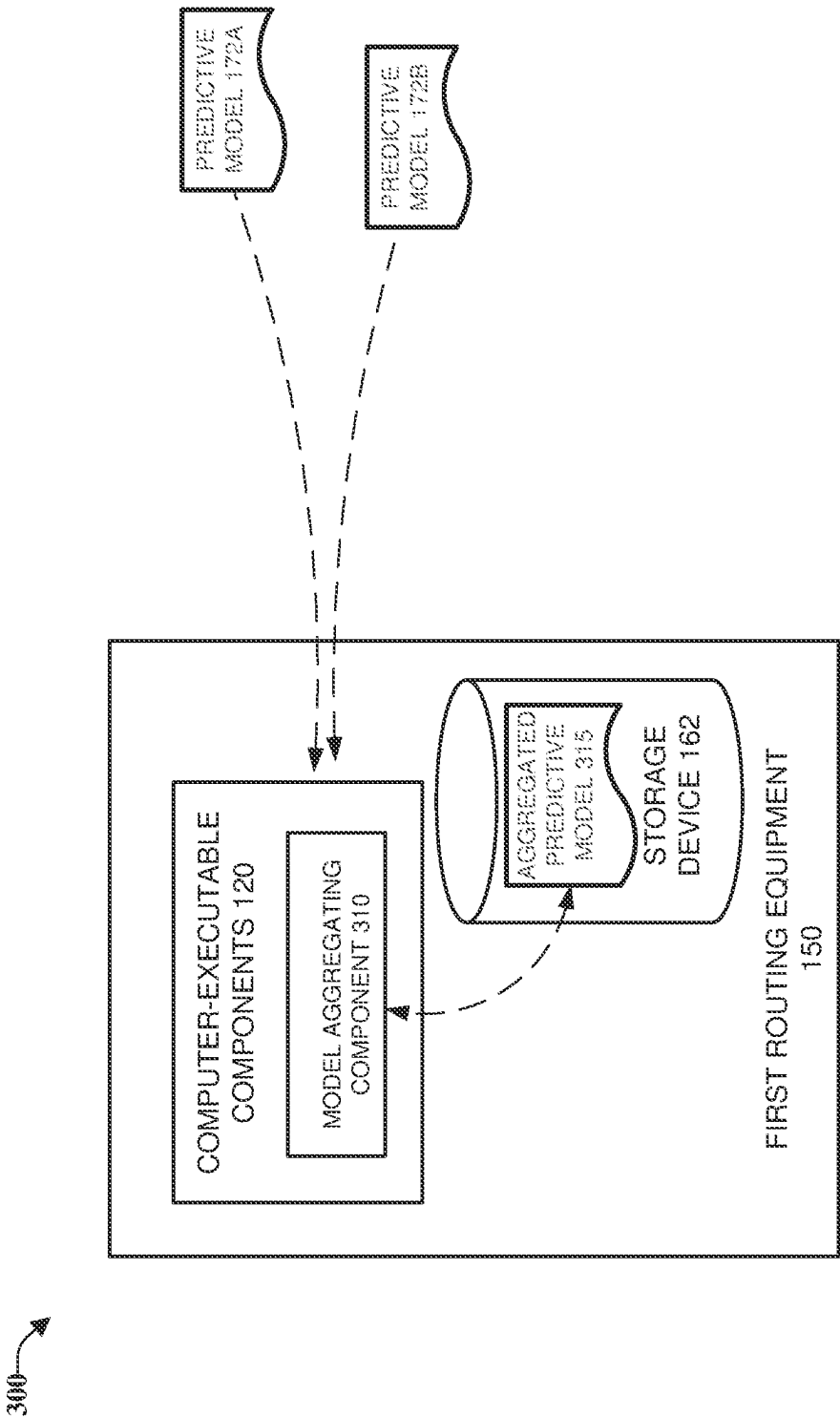


FIG. 3

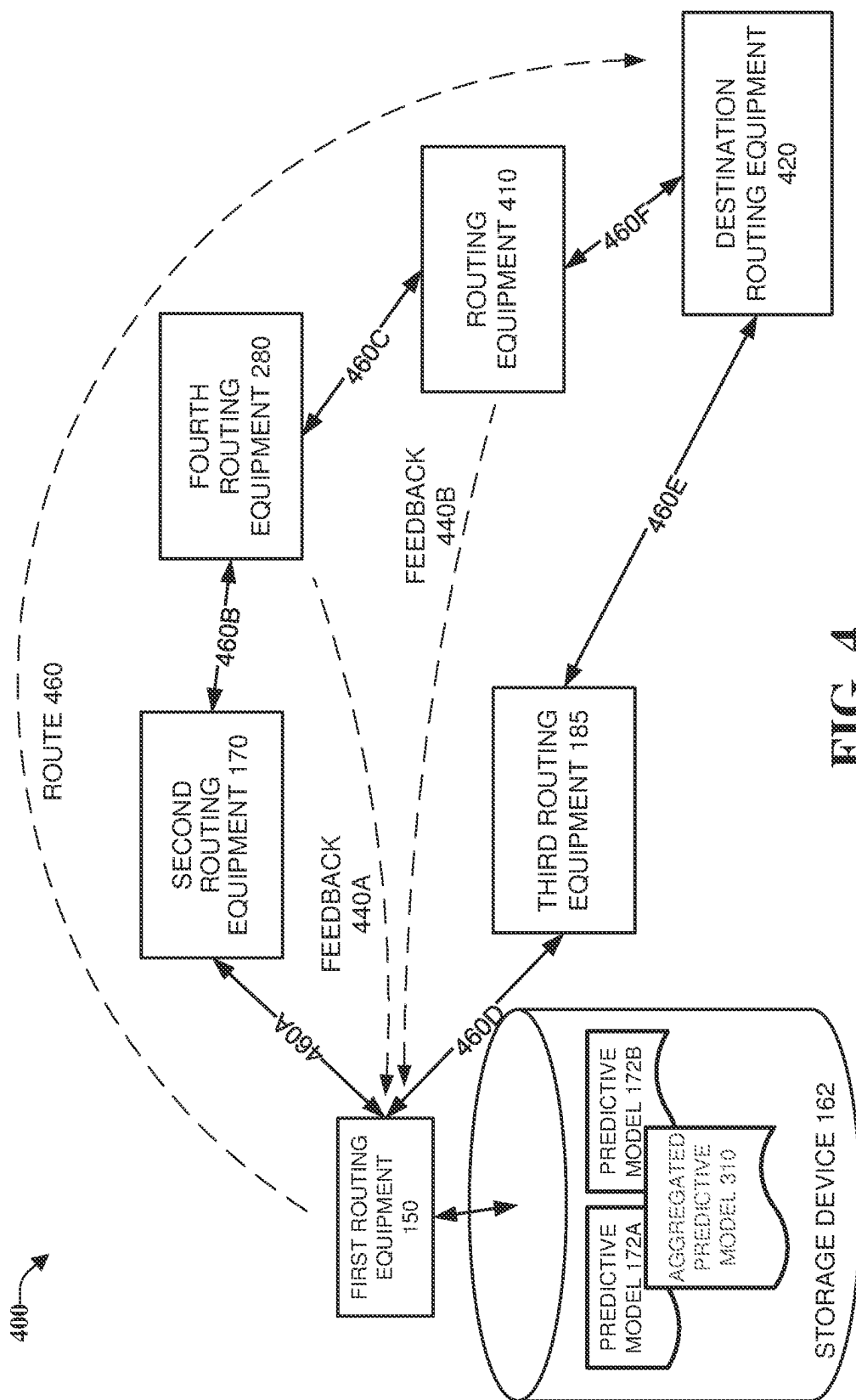


FIG. 4

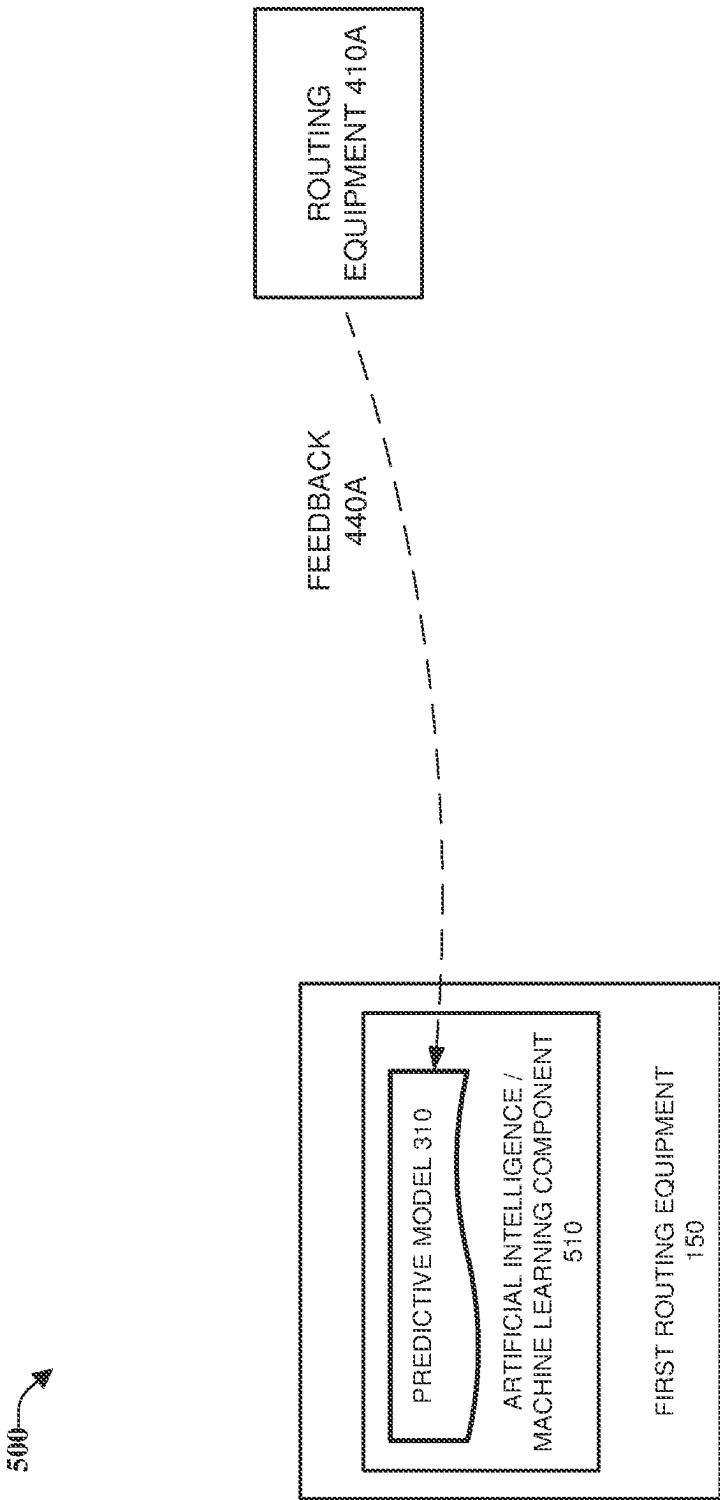
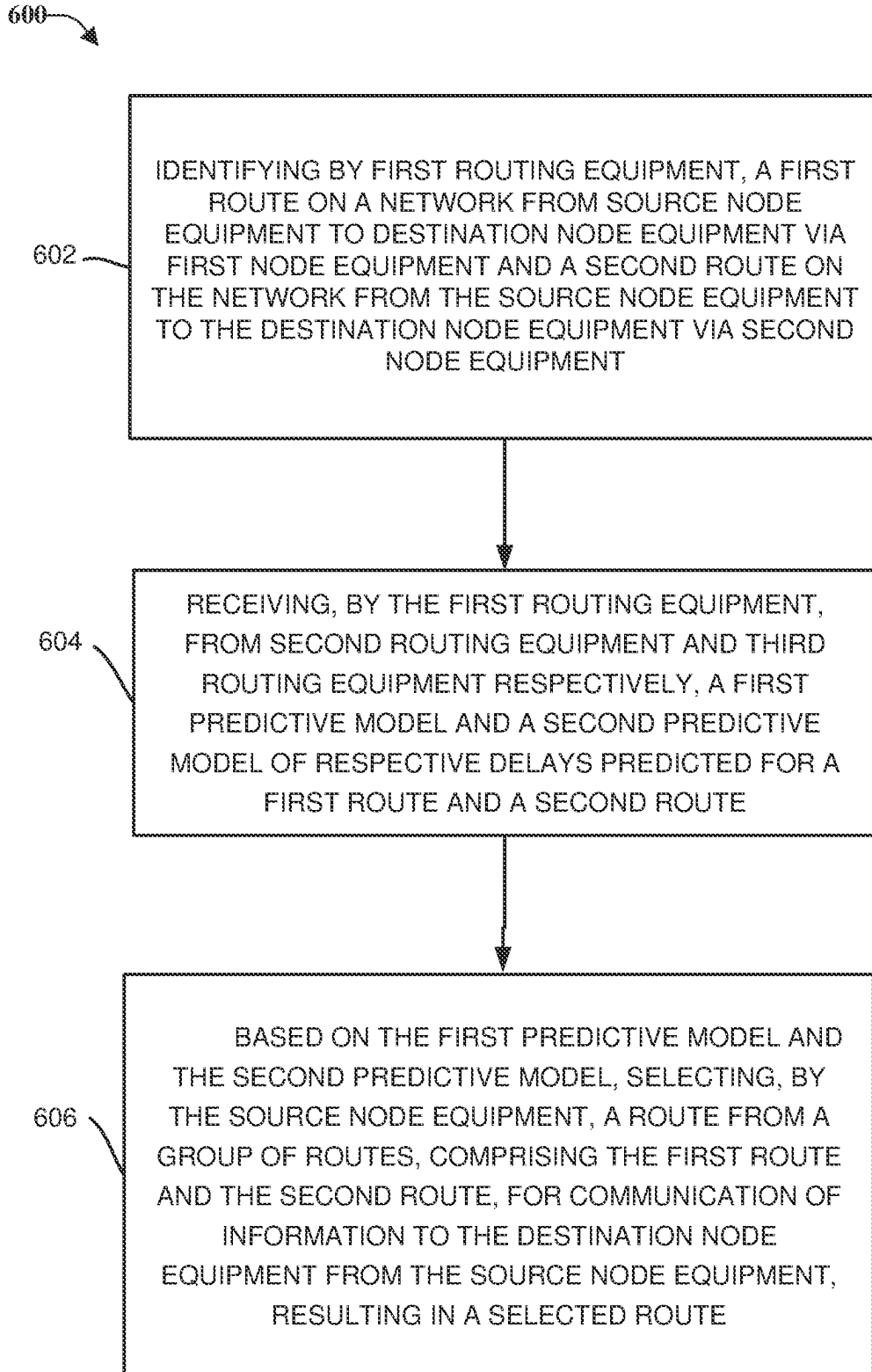
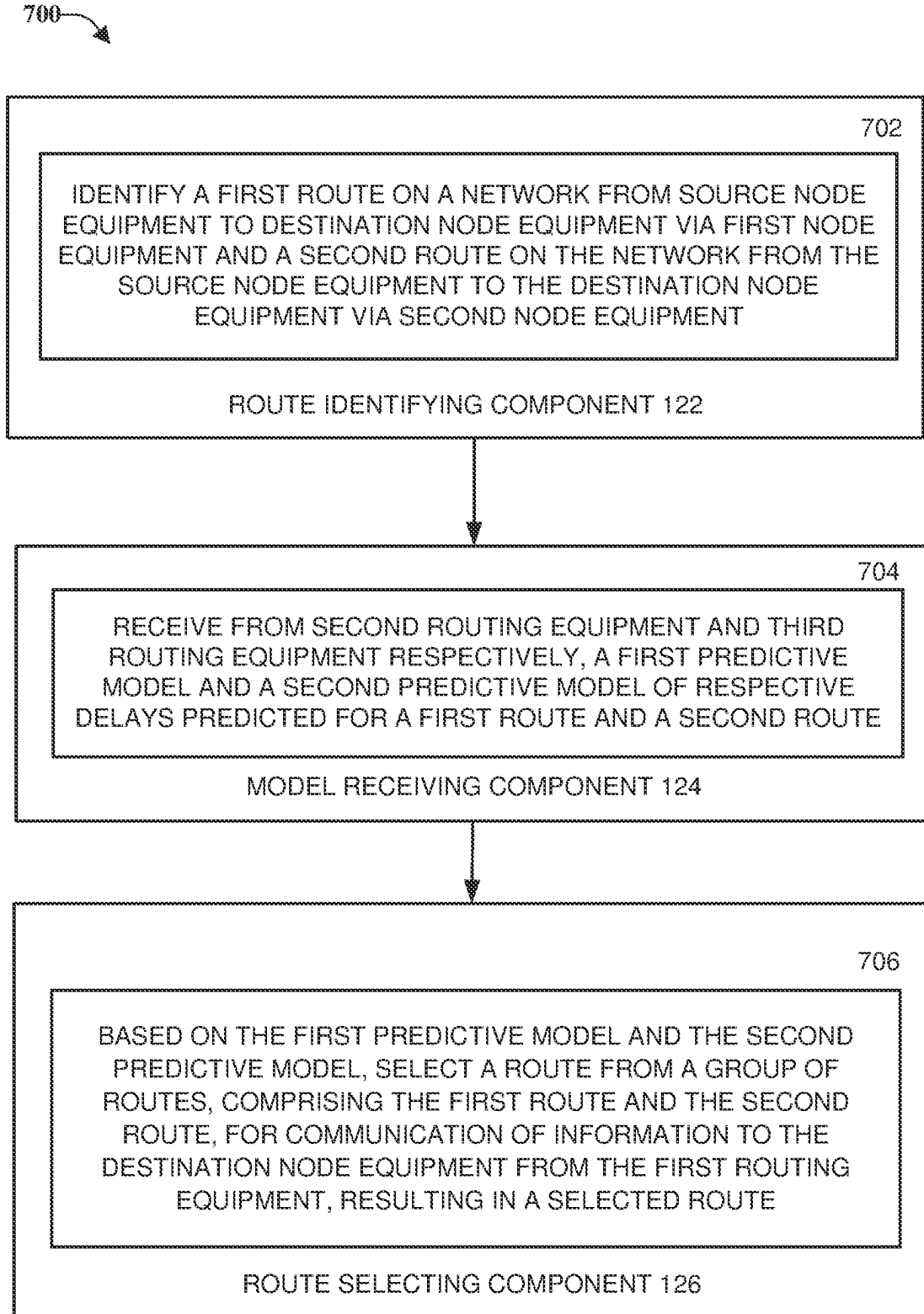


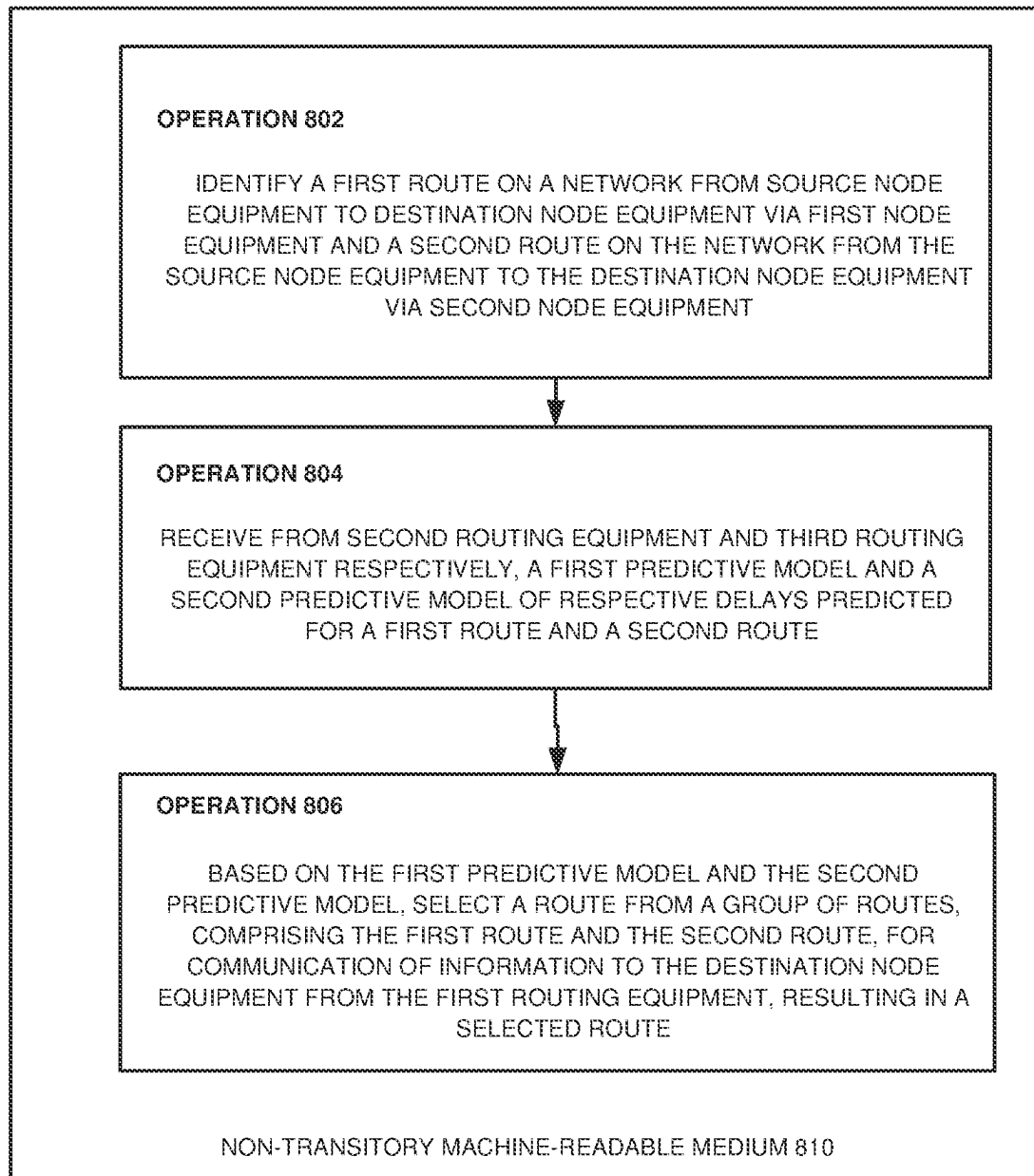
FIG. 5

**FIG. 6**



**FIG. 7**

800

**FIG. 8**

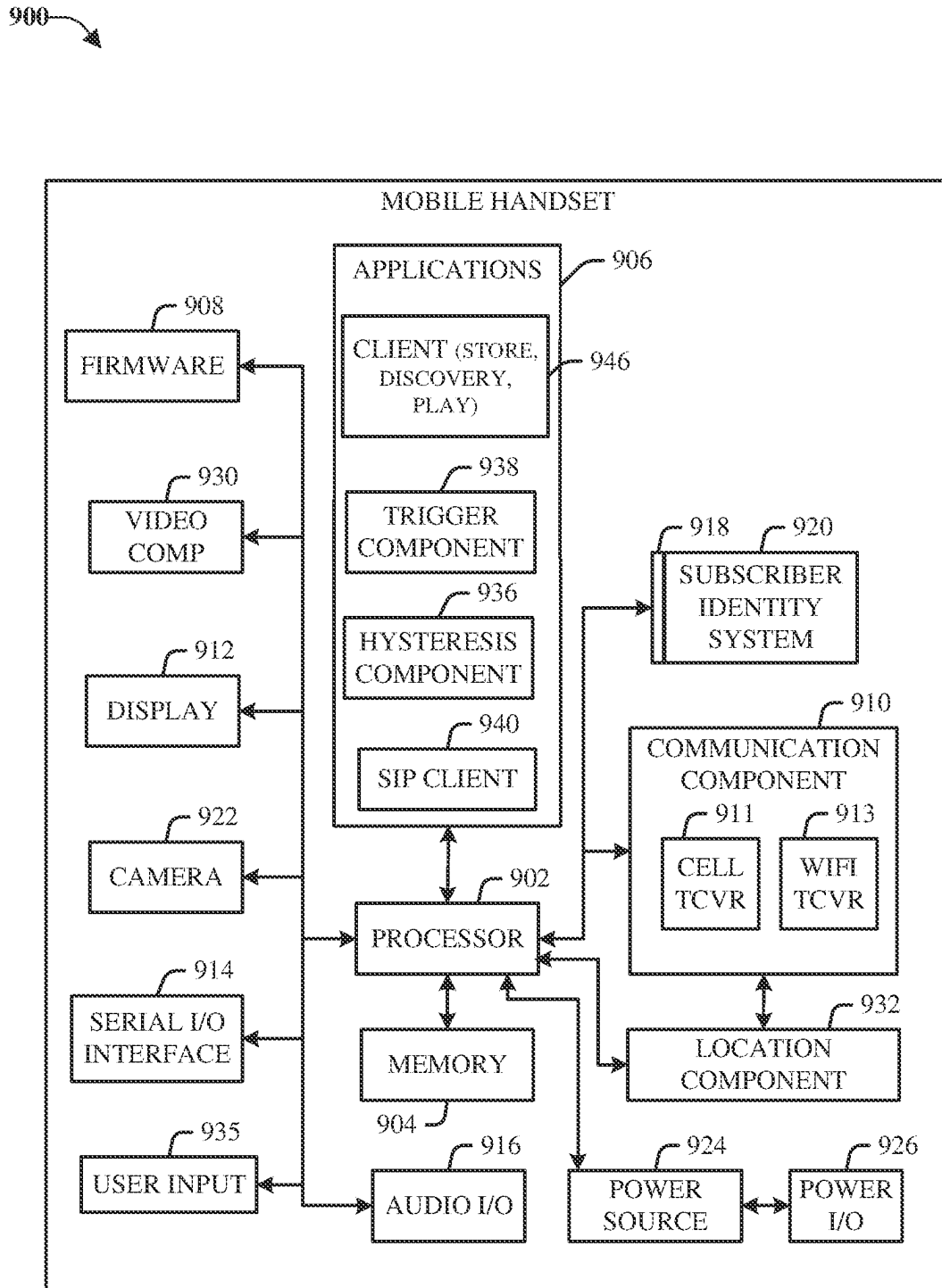


FIG. 9

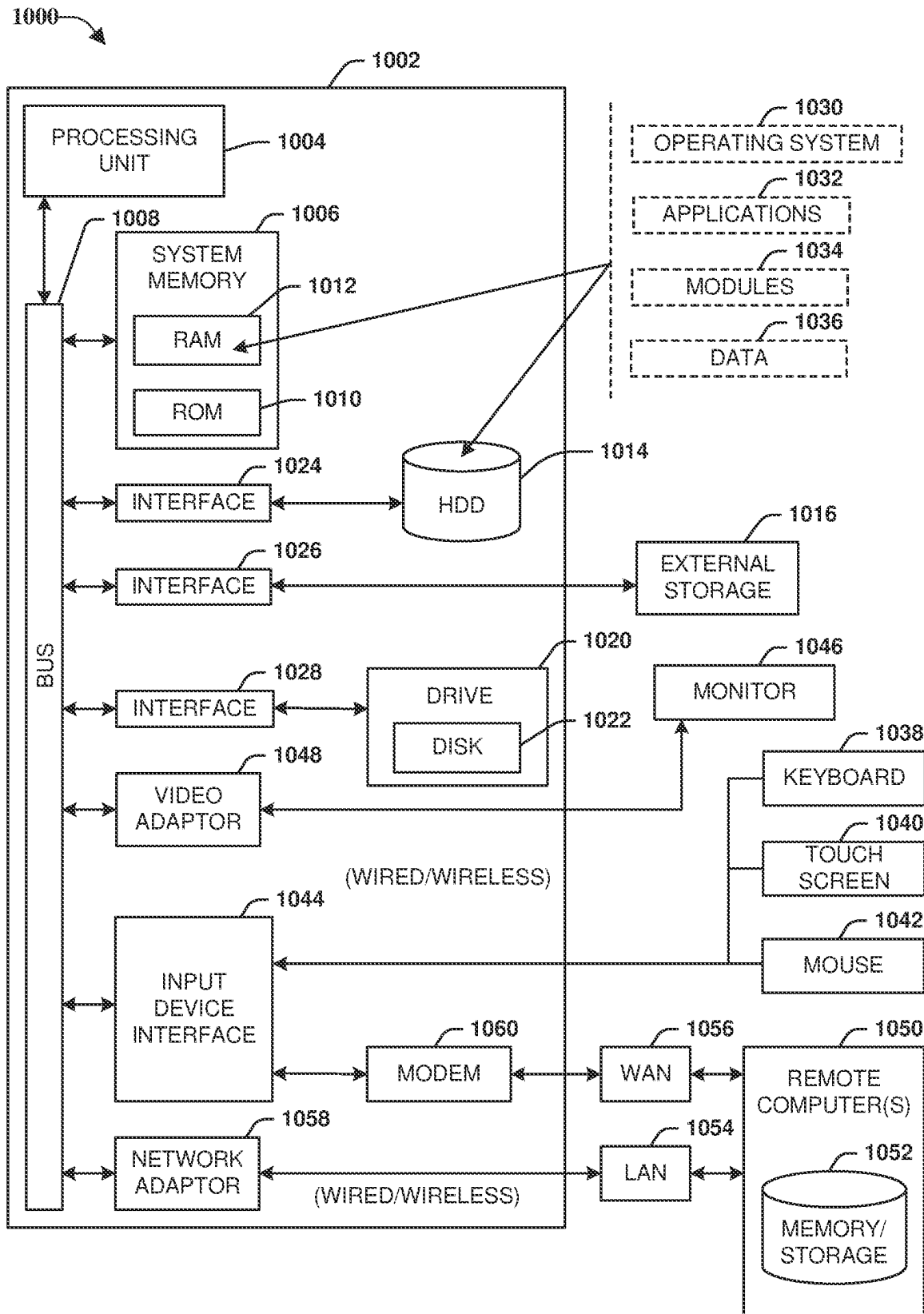


FIG. 10

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## SELECTING NETWORK ROUTES BASED ON AGGREGATING MODELS THAT PREDICT NODE ROUTING PERFORMANCE

### TECHNICAL FIELD

The subject application is related to different approaches to handling communication in networked computer systems and, for example, to selecting network routes based on predicted performance.

### BACKGROUND

As network transmission speeds continue to increase, delays caused by individual nodes of a communication route have increased in significance. Delays that were insignificant in comparison to past transmission speeds have significantly increased in importance with other parts of the network having much higher throughput. Unlike Bandwidth, which in some circumstances can be increased by adding more fiber/circuits, delays associated with routing nodes can be limited by the physical capacity of the routing nodes, e.g., processing speed of router interfaces generally cannot be easily increased to mitigate problematic delays.

These problems can become even more significant with the increase in the number of possible routes available for communication. In some circumstances, by the time routing conditions of route nodes are received at upstream nodes selecting routes, the conditions have changed, and are no longer as useful for selecting routes.

### BRIEF DESCRIPTION OF THE DRAWINGS

The technology described herein is illustrated by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:

FIG. 1 is an architecture diagram of an example system that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments.

FIG. 2 is a diagram of a non-limiting example system that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments.

FIG. 3 is a diagram of a non-limiting example system that can facilitate aggregating models that can predict node routing performance, in accordance with one or more embodiments.

FIG. 4 depicts an example collection of interconnected nodes that can have some route selections performed by one or more embodiments described herein.

FIG. 5 is a diagram of a non-limiting example system that can facilitate selecting network routes based on predictive models generated and maintained by employing artificial intelligence/machine learning (AI/ML) approaches, in accordance with one or more embodiments.

FIG. 6 illustrates an example method that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments.

FIG. 7 depicts a system that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments.

FIG. 8 depicts an example non-transitory machine-readable medium that can include executable instructions that,

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when executed by a processor of a system, facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments described above.

FIG. 9 illustrates an example block diagram of an example mobile handset operable to engage in a system architecture that can facilitate processes described herein, in accordance with one or more embodiments.

FIG. 10 illustrates an example block diagram of an example computer operable to engage in a system architecture that can facilitate processes described herein, in accordance with one or more embodiments.

### DETAILED DESCRIPTION

Generally speaking, one or more embodiments can facilitate selecting network routes based on aggregating models that can predict routing performance. In addition, one or more embodiments described herein can be directed towards a multi-connectivity framework that supports the operation of new radio (NR, sometimes referred to as 5G). As will be understood, one or more embodiments can allow an integration of user devices with network assistance, by supporting control and mobility functionality on cellular links (e.g., long term evolution (LTE) or NR). One or more embodiments can provide benefits including, system robustness, reduced overhead, and global resource management, while facilitating direct communication links via a NR sidelink.

It should be understood that any of the examples and terms used herein are non-limiting. For instance, while examples are generally directed to non-standalone operation where the NR backhaul links are operating on millimeter wave (mmWave) bands and the control plane links are operating on sub-6 GHz LTE bands, it should be understood that it is straightforward to extend the technology described herein to scenarios in which the sub-6 GHz anchor carrier providing control plane functionality could also be based on NR. As such, any of the examples herein are non-limiting examples, any of the embodiments, aspects, concepts, structures, functionalities or examples described herein are non-limiting, and the technology may be used in various ways that provide benefits and advantages in radio communications in general.

In some embodiments the non-limiting terms “signal propagation equipment” or simply “propagation equipment,” “radio network node” or simply “network node,” “radio network device,” “network device,” and access elements are used herein. These terms may be used interchangeably, and refer to any type of network node that can serve user equipment and/or be connected to other network node or network element or any radio node from where user equipment can receive a signal. Examples of radio network node include, but are not limited to, base stations (BS), multi-standard radio (MSR) nodes such as MSR BS, gNodeB, eNode B, network controllers, radio network controllers (RNC), base station controllers (BSC), relay, donor node controlling relay, base transceiver stations (BTS), access points (AP), transmission points, transmission nodes, remote radio units (RRU) (also termed radio units herein), remote radio heads (RRH), and nodes in distributed antenna system (DAS). Additional types of nodes are also discussed with embodiments below, e.g., donor node equipment and relay node equipment, an example use of these being in a network with an integrated access backhaul network topology.

In some embodiments, the non-limiting term user equipment (UE) is used. This term can refer to any type of

wireless device that can communicate with a radio network node in a cellular or mobile communication system. Examples of UEs include, but are not limited to, a target device, device to device (D2D) user equipment, machine type user equipment, user equipment capable of machine to machine (M2M) communication, PDAs, tablets, mobile terminals, smart phones, laptop embedded equipped (LEE), laptop mounted equipment (LME), USB dongles, and other equipment that can have similar connectivity. Example UEs are described further with FIGS. 9 and 10 below. Some embodiments are described in particular for 5G new radio systems. The embodiments are however applicable to any radio access technology (RAT) or multi-RAT system where the UEs operate using multiple carriers, e.g., LTE.

The computer processing systems, computer-implemented methods, apparatus and/or computer program products described herein employ hardware and/or software to solve problems that are highly technical in nature (e.g., processing multiple predictive models of network performance), that are not abstract and cannot be performed as a set of mental acts by a human. For example, a human, or even a plurality of humans, cannot efficiently and quickly analyze the relevant data with the same level of accuracy and/or efficiency as the various embodiments described herein.

Aspects of the subject disclosure will now be described more fully hereinafter with reference to the accompanying drawings in which example components, graphs and selected operations are shown. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the various embodiments. For example, some embodiments described can facilitate selecting network routes based on aggregating models that can predict routing performance. Different examples that describe these aspects are included with the description of FIGS. 1-10 below. It should be noted that the subject disclosure may be embodied in many different forms and should not be construed as limited to this example or other examples set forth herein.

FIG. 1 is an architecture diagram of an example system 100 that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted. It should be noted that, although many examples herein discuss user equipment with one additional network identifier (e.g., dual-provisioned user equipment), one having skill in the relevant art(s), given the description herein would appreciate that the approaches can also apply to any number of network identifiers associated with a user equipment.

As depicted, system 100 can include first routing equipment 150 communicatively coupled via routes 126A-B to second routing equipment 170 and third routing equipment 185 via network 190, respectively. As depicted, routing equipment can receive predictive models 172A-B from second routing equipment 170 and third routing equipment 185, respectively. It should be noted that, as discussed herein, first routing equipment 150 can also be termed a node, router or device, without deviating from the spirit of embodiments described herein. Further, because whether first routing equipment 150 receives communications (e.g., transmission control protocol/internet protocol (TCP/IP) packets to be relayed to a destination node) from another 'upstream' routing device, or is the originator of a communication, first routing equipment 150 can be termed herein as

a source node, e.g., as the node that is currently determining to which 'downstream' routing device to relay the communication.

As depicted, first routing equipment 150 can include computer executable components 120, processor 160, storage device 162, and memory 165. Computer executable components 120 can include route identifying component 122, model receiving component 124, route selecting component 126, and other components described or suggested by different embodiments described herein, that can improve the operation of system 100.

Further to the above, it should be appreciated that these components, as well as aspects of the embodiments of the subject disclosure depicted in this figure and various figures disclosed herein, are for illustration only, and as such, the architecture of such embodiments are not limited to the systems, devices, and/or components depicted therein. For example, in some embodiments, first routing equipment 150 can further comprise various computer and/or computing-based elements described herein with reference to mobile handset 900 of FIG. 9, and operating environment 1000 of FIG. 10. For example, one or more of the different functions of network equipment can be divided among various equipment, including, but not limited to, including equipment at a central node global control located on the core Network, e.g., mobile edge computing (MEC), self-organized networks (SON), or RAN intelligent controller (RIC) network equipment.

In some embodiments, memory 165 can comprise volatile memory (e.g., random access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), etc.) and/or non-volatile memory (e.g., read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), etc.) that can employ one or more memory architectures. Further examples of memory 165 are described below with reference to system memory 1006 and FIG. 10. Such examples of memory 165 can be employed to implement any embodiments of the subject disclosure.

According to multiple embodiments, storage device 162 can include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, solid state drive (SSD) or other solid-state storage technology, Compact Disk Read Only Memory (CD ROM), digital video disk (DVD), blu-ray disk, or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computer.

According to multiple embodiments, processor 160 can comprise one or more processors and/or electronic circuitry that can implement one or more computer and/or machine readable, writable, and/or executable components and/or instructions that can be stored on memory 165. For example, processor 160 can perform various operations that can be specified by such computer and/or machine readable, writable, and/or executable components and/or instructions including, but not limited to, logic, control, input/output (I/O), arithmetic, and/or the like. In some embodiments, processor 160 can comprise one or more components including, but not limited to, a central processing unit, a multi-core processor, a microprocessor, dual microprocessors, a micro-controller, a system on a chip (SOC), an array processor, a vector processor, and other types of processors. Further examples of processor 160 are described below with reference to processing unit 1004 of FIG. 10. Such examples of

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processor **160** can be employed to implement any embodiments of the subject disclosure.

In one or more embodiments, computer executable components **120** can be used in connection with implementing one or more of the systems, devices, components, and/or computer-implemented operations shown and described in connection with FIG. **1** or other figures disclosed herein. For example, in one or more embodiments, computer executable components **120** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining route identifying component **122**. As discussed with FIGS. **2-4** below, route identifying component **122** can, in accordance with one or more embodiments, identify a first route on a network from the source node equipment to destination node equipment via first node equipment and a second route on the network from the source node equipment to the destination node equipment via second node equipment. For example, one or more embodiments of route identifying component **122** can identify routes **126A-B** via network **190** from first routing equipment **150** to second routing equipment **170** and third routing equipment **185** via network **190**, respectively. Additional details regarding route selection by route identifying component **122** are provided with the discussion of FIGS. **2** and **4** below.

Further, in another example, in one or more embodiments, computer executable components **120** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining model receiving component **124**. As discussed with FIGS. **3-4** below, model receiving component **124** can, in accordance with one or more embodiments, receive from other node equipment, predictive models of respective delays predicted for different available routes. For example, in one or more embodiments, receive, by the first routing equipment **150**, from the second routing equipment **170** and the third routing equipment **185** respectively, a first predictive model **172A** and a second predictive model **172B** of respective delays predicted for the first route and second route. One having skill in the relevant art(s), given the description herein, appreciates that different approaches can be used to implement the predictive models, including various machine learning approaches described with FIG. **5** below.

Additional details regarding the generation of predictive models and receipt and use by model receiving component **124** are provided with the discussion of FIGS. **2** and **4** below.

In yet another example, computer executable components **120** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining route selecting component **126**. As discussed herein, route selecting component **126** can, based on the first predictive model **172A** and the second predictive model **172B**, select a route from a group of routes, e.g., based on predictive model **172A**, route selecting component **126** can select route **126A** from the group of routes **126A-B**.

FIG. **2** is a diagram of a non-limiting example system **200** that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted.

As depicted, system **200** can include second routing equipment **170** connected to first routing equipment **150** and fourth routing equipment **280** via route **126A** and route **226** via network **190**. Second routing equipment **170** can include memory **165** that can store one or more computer and/or machine readable, writable, and/or executable components

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and/or instructions **220** that, when respectively executed by processor **160**, can facilitate performance of operations defined by the executable component(s) and/or instruction(s).

Generally, applications (e.g., computer executable components **220**) can include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. In system **200**, computer executable components **220** can include network condition estimating component **212**, receiving component **214**, communicating component **216**, and other components described or suggested by different embodiments described herein that can improve the operation of system **200**. It should be appreciated that these components, as well as aspects of the embodiments of the subject disclosure depicted in this figure and various figures disclosed herein, are for illustration only, and as such, the architecture of such embodiments are not limited to the systems, devices, and/or components depicted therein. For example, in some embodiments, one or more of the routing equipment discussed herein can further comprise various computer and/or computing-based elements described herein with reference to mobile handset **900** of FIG. **9** and operating environment **1000** described with FIG. **10**.

For example, in one or more embodiments, computer executable components **220** can be used in connection with implementing one or more of the systems, devices, components, and/or computer-implemented operations shown and described in connection with FIG. **2** or other figures disclosed herein. For example, in one or more embodiments, computer executable components **220** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining network condition estimating component **212**. As discussed with FIGS. **4-5** below, in one or more embodiments, network condition estimating component **212** can, communicate, to first routing equipment, a first model describing a delay predicted to be caused to a future communication by the future communication being transited via the second routing equipment. For example, in one or more embodiments, network condition estimating component **212** can communicate predictive model **172A** to first routing equipment **150**, with this predictive model **172A** describing the network conditions (e.g., delay) predicted to be caused to a future communication by the future communication being transited via the second routing equipment **170**.

In another example, in one or more embodiments, computer executable components **220** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining, receiving component **214**. As discussed with FIGS. **4-5** below, receiving component **214** can, in accordance with one or more embodiments, receive, from the first routing equipment, a current communication for transit via second routing equipment **170** to destination equipment (e.g., fourth routing equipment **280**), with second routing equipment **170** being selected by first routing equipment **150** based on the first model, and the second model, other than the first model, describing respective predicted delays from other routing equipment other than the first routing and second routing equipment **185**.

In another example, in one or more embodiments, computer executable components **220** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining, communicating component **216**. As discussed with FIGS. **4-5** below, communicating component **216** can, in accordance with one or more embodiments, based on predictive model **172A**, can relay

the current communication to second routing equipment **170** to transit the current communication to destination equipment, e.g., equipment corresponding to the destination address of a packet forwarded via TCP/IP protocol to a next selected network segment (router, node).

FIGS. **3** and **4** are diagrams that illustrate different approaches that can be employed by one or more embodiments to route communications, including routing with aggregated predictive models described herein. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted.

As depicted in FIG. **3**, system **300** can include first routing equipment **150** receiving predictive models **172A-B**, e.g., from second routing equipment **170** and third routing equipment **185**, respectively. In this depiction of first routing equipment **150**, computer executable components **120** further included model aggregating component **310** communicatively coupled to storage device **162**. Storage device **162** is depicted in a non-limiting manner as an example component to store aggregated predictive model **315**, e.g., generated and maintained by model aggregating component **310**.

FIG. **4** depicts an example **400** collection of interconnected nodes that can have some route selections performed by one or more embodiments described herein. Example **400** includes first routing component **150**, second routing equipment **170**, third routing equipment **185**, fourth routing equipment **280**, routing equipment **410**, and destination routing equipment **420**. First routing component **150** is communicatively coupled to storage device **162**, which stores aggregated predictive model **315** being composed of an aggregate of predictive models **172A-B**.

In an alternate embodiment to some of the embodiments above, as depicted in FIGS. **3** and **4**, when first routing equipment **150** is selecting between routes **460A** and **460D** to second routing equipment **170** and third routing equipment **185** respectively, in addition to utilizing predictive models **172A-B**, first routing equipment **150** can use model aggregating component **310** to predictive models **172A-B** into aggregated predictive model **315** stored, for example, in storage device **162**. One having skill in the relevant art(s), given the description herein, appreciates that different approaches can be used to aggregate predictive models **172A-B**, including approaches discussed below with FIG. **5** that utilize neural networks.

It should be noted that, in the network shown in FIG. **4**, some or all of the routing equipment depicted can use a component similar to model aggregating component **310** to aggregate predictive models from connected nodes. In some implementations, these aggregated models can be passed to other connected nodes, where they are further distributed. Thus, in an example depicted destination routing equipment **420** can generate a predictive model of network conditions (e.g., based on models).

In one approach, first routing equipment **150** can generate a local predictive model of available network routes, e.g., using a component similar to network condition estimating component **212** described with FIG. **2** above. This generated predictive model can be provided using connections **460C** and **460E** to routing equipment **410** and third routing equipment **185**, respectively, where this model can be aggregated with the present predictive model at this equipment. By this process, the aggregated predictive models from second routing equipment **170** and third routing equipment **185** can reach first routing equipment **150**, where the resulting aggregate-

gated predictive model **315** can predict conditions in both the **460A**, **460B**, **460C**, and **460F** route and the **460D**, **460E** route.

In additional embodiment, after a predictive model (e.g., aggregated predictive model **315**) is used to select a route (e.g., route **460**), routing components along the route can provide feedback **440A-B** to the routing equipment that selected the route (e.g., first routing equipment **150**). This feedback can be used to incrementally update aspects of different models, e.g., to increase the accuracy of predictions provided.

FIG. **5** is a diagram of a non-limiting example system **500** that can facilitate selecting network routes based on predictive models generated and maintained by employing AI/ML approaches, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted.

As depicted, system **500** can comprise AI/ML components **510** to generate and maintaining aggregated predictive model **315** to facilitate selecting network routes as described herein. In one or more embodiments, AI/ML components **510** can comprise an artificial neural network (ANN), e.g., initially trained and subsequently updated by predictive models received from other routers, and feedback **440A-B** provided in response to previous route selections. Example inputs that can be used to train AI/ML components **510** can include historical network node performance, and feedback **440A-B** from specific route selections.

In certain embodiments, different functions of AI/ML component **510** can be facilitated based on principles of AI that include, but are not limited to, classifications, correlations, inferences and/or expressions, with for example, AI/ML component **510** employing approaches that include, but are not limited to, expert systems, fuzzy logic, state vector machines (SVMs), Hidden Markov Models (HMMs), greedy search algorithms, rule-based systems, Bayesian models (e.g., Bayesian networks), non-linear training techniques, data fusion, and utility-based analytical systems. Additional implementations can include ensemble ML algorithms/methods, including deep neural networks (DNN), reinforcement learning (RL), and long short-term memory (LSTM) networks.

FIG. **6** illustrates an example method **600** that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted.

At **602**, method **600** can include identifying, by source node equipment comprising a processor, a first route on a network from the source node equipment to destination node equipment via first node equipment and a second route on the network from the source node equipment to the destination node equipment via second node equipment. For example, in one or more embodiments, route identifying component **122** can identify a first route **126A** on a network **190** from the source node equipment to destination node equipment via first node equipment and a second route **126B** on the network from the source node equipment to the destination node equipment via second node equipment.

At **604**, method **600** can include, receiving, by the source node equipment, from the first and the second node equipment respectively, a first predictive model and a second predictive model of respective delays predicted for the first route and second route. For example, in one or more embodiments, model receiving component **124** can receive from the first and the second node equipment respectively,



a first predictive model **172A** and a second predictive model **172B** of respective delays predicted for the first route **126A** and second route **126B**.

At **606**, method **600** can include, based on the first predictive model and the second predictive model, employing route selecting component **126** to select, by the source node equipment, a route from a group of routes, comprising the first route and the second route, for communication of information to the destination node equipment.

FIG. **7** depicts a system **700** that can facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted. As depicted, system **700** can include route identifying component **122**, model receiving component **124**, route selecting component **126**, and other components described or suggested by different embodiments described herein, that can improve the operation of system **700**.

In an example, component **702** can include the functions of route identifying component **122**, supported by the other layers of system **700**. For example, in an embodiment, component **702** can identify a first route on a network from the source node equipment to destination node equipment via first node equipment and a second route on the network from the source node equipment to the destination node equipment via second node equipment.

In this and other examples, component **704** can include the functions of model receiving component **124**, supported by the other layers of system **700**. Continuing this example, in one or more embodiments, component **704** can receive, from the first and the second node equipment respectively, a first predictive model and a second predictive model of respective delays predicted for the first route and second route.

In an additional example, component **706** can include the functions of route selecting component **126**, supported by the other layers of system **700**. For example, component **706** can employ route selecting component **126** to select, by the source node equipment, a route from a group of routes, comprising the first route and the second route, for communication of information to the destination node equipment.

FIG. **8** depicts an example **800** non-transitory machine-readable medium **810** that can include executable instructions that, when executed by a processor of a system, facilitate selecting network routes based on aggregating models that can predict routing performance, in accordance with one or more embodiments described above. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted. As depicted, non-transitory machine-readable medium **810** includes executable instructions that can facilitate performance of operations **802-808**.

In one or more embodiments, the operations can include operation **802** that can include identifying, by source node equipment comprising a processor, a first route on a network from the source node equipment to destination node equipment via first node equipment and a second route on the network from the source node equipment to the destination node equipment via second node equipment.

Operations can further include operation **804**, that can receive from the first and the second node equipment respectively, a first predictive model and a second predictive model of respective delays predicted for the first route and second route. In one or more embodiments, the operations can further include operation **806** that can, based on the first predictive model and the second predictive model, select, by

the source node equipment, a route from a group of routes, comprising the first route and the second route, for communication of information to the destination node equipment from the source node equipment, resulting in a selected route.

FIG. **9** illustrates an example block diagram of an example mobile handset **900** operable to engage in a system architecture that facilitates wireless communications according to one or more embodiments described herein. Although a mobile handset is illustrated herein, it will be understood that other devices can be a mobile device, and that the mobile handset is merely illustrated to provide context for the embodiments of the various embodiments described herein. The following discussion is intended to provide a brief, general description of an example of a suitable environment in which the various embodiments can be implemented. While the description includes a general context of computer-executable instructions embodied on a machine-readable storage medium, those skilled in the art will recognize that the embodiments also can be implemented in combination with other program modules and/or as a combination of hardware and software.

Generally, applications (e.g., program modules) can include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods described herein can be practiced with other system configurations, including single-processor or multiprocessor systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

A computing device can typically include a variety of machine-readable media. Machine-readable media can be any available media that can be accessed by the computer and includes both volatile and non-volatile media, removable and non-removable media. By way of example and not limitation, computer-readable media can comprise computer storage media and communication media. Computer storage media can include volatile and/or non-volatile media, removable and/or non-removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, program modules, or other data. Computer storage media can include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, solid state drive (SSD) or other solid-state storage technology, Compact Disk Read Only Memory (CD ROM), digital video disk (DVD), Blu-ray disk, or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computer. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

Communication media typically embodies computer-readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the

signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of the any of the above should also be included within the scope of computer-readable media.

The handset includes a processor **902** for controlling and processing all onboard operations and functions. A memory **904** interfaces to the processor **902** for storage of data and one or more applications **906** (e.g., a video player software, user feedback component software, etc.). Other applications can include voice recognition of predetermined voice commands that facilitate initiation of the user feedback signals. The applications **906** can be stored in the memory **904** and/or in a firmware **908**, and executed by the processor **902** from either or both the memory **904** or/and the firmware **908**. The firmware **908** can also store startup code for execution in initializing the handset **900**. A communications component **910** interfaces to the processor **902** to facilitate wired/wireless communication with external systems, e.g., cellular networks, VoIP networks, and so on. Here, the communications component **910** can also include a suitable cellular transceiver **911** (e.g., a GSM transceiver) and/or an unlicensed transceiver **913** (e.g., Wi-Fi, WiMax) for corresponding signal communications. The handset **900** can be a device such as a cellular telephone, a PDA with mobile communications capabilities, and messaging-centric devices. The communications component **910** also facilitates communications reception from terrestrial radio networks (e.g., broadcast), digital satellite radio networks, and Internet-based radio services networks.

The handset **900** includes a display **912** for displaying text, images, video, telephony functions (e.g., a Caller ID function), setup functions, and for user input. For example, the display **912** can also be referred to as a "screen" that can accommodate the presentation of multimedia content (e.g., music metadata, messages, wallpaper, graphics, etc.). The display **912** can also display videos and can facilitate the generation, editing and sharing of video quotes. A serial I/O interface **914** is provided in communication with the processor **902** to facilitate wired and/or wireless serial communications (e.g., USB, and/or IEEE 1294) through a hardware connection, and other serial input devices (e.g., a keyboard, keypad, and mouse). This supports updating and troubleshooting the handset **900**, for example. Audio capabilities are provided with an audio I/O component **916**, which can include a speaker for the output of audio signals related to, for example, indication that the user pressed the proper key or key combination to initiate the user feedback signal. The audio I/O component **916** also facilitates the input of audio signals through a microphone to record data and/or telephony voice data, and for inputting voice signals for telephone conversations.

The handset **900** can include a slot interface **918** for accommodating a SIC (Subscriber Identity Component) in the form factor of a card SIM or universal SIM **920**, and interfacing the SIM card **920** with the processor **902**. However, it is to be appreciated that the SIM card **920** can be manufactured into the handset **900**, and updated by downloading data and software.

The handset **900** can process IP data traffic through the communications component **910** to accommodate IP traffic from an IP network such as, for example, the Internet, a corporate intranet, a home network, a person area network, etc., through an ISP or broadband cable provider. Thus, VoIP

traffic can be utilized by the handset **900** and IP-based multimedia content can be received in either an encoded or a decoded format.

A video processing component **922** (e.g., a camera) can be provided for decoding encoded multimedia content. The video processing component **922** can aid in facilitating the generation, editing, and sharing of video quotes. The handset **900** also includes a power source **924** in the form of batteries and/or an AC power subsystem, which power source **924** can interface to an external power system or charging equipment (not shown) by a power I/O component **926**.

The handset **900** can also include a video component **930** for processing video content received and, for recording and transmitting video content. For example, the video component **930** can facilitate the generation, editing and sharing of video quotes. A location tracking component **932** facilitates geographically locating the handset **900**. As described hereinabove, this can occur when the user initiates the feedback signal automatically or manually. A user input component **934** facilitates the user initiating the quality feedback signal. The user input component **934** can also facilitate the generation, editing and sharing of video quotes. The user input component **934** can include such conventional input device technologies such as a keypad, keyboard, mouse, stylus pen, and/or touch screen, for example.

Referring again to the applications **906**, a hysteresis component **936** facilitates the analysis and processing of hysteresis data, which is utilized to determine when to associate with the access point. A software trigger component **938** can be provided that facilitates triggering of the hysteresis component **936** when the Wi-Fi transceiver **913** detects the beacon of the access point. A SIP client **940** enables the handset **900** to support SIP protocols and register the subscriber with the SIP registrar server. The applications **906** can also include a client **942** that provides at least the capability of discovery, play and store of multimedia content, for example, music.

The handset **900**, as indicated above related to the communications component **910**, includes an indoor network radio transceiver **913** (e.g., Wi-Fi transceiver). This function supports the indoor radio link, such as IEEE 802.11, for the dual-mode GSM handset **900**. The handset **900** can accommodate at least satellite radio services through a handset that can combine wireless voice and digital radio chipsets into a single handheld device.

Network **190** can employ various cellular systems, technologies, and modulation schemes to facilitate wireless radio communications between devices. While example embodiments include use of 5G new radio (NR) systems, one or more embodiments discussed herein can be applicable to any radio access technology (RAT) or multi-RAT system, including where user equipment operate using multiple carriers, e.g., LTE FDD/TDD, GSM/GERAN, CDMA2000, etc. For example, wireless communication system **200** can operate in accordance with global system for mobile communications (GSM), universal mobile telecommunications service (UMTS), long term evolution (LTE), LTE frequency division duplexing (LTE FDD, LTE time division duplexing (TDD), high speed packet access (HSPA), code division multiple access (CDMA), wideband CDMA (WCDMA), CDMA2000, time division multiple access (TDMA), frequency division multiple access (FDMA), multi-carrier code division multiple access (MC-CDMA), single-carrier code division multiple access (SC-CDMA), single-carrier FDMA (SC-FDMA), orthogonal frequency division multiplexing (OFDM), discrete Fourier transform spread OFDM (DFT-spread OFDM) single carrier

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FDMA (SC-FDMA), Filter bank based multi-carrier (FBMC), zero tail DFT-spread-OFDM (ZT DFT-s-OFDM), generalized frequency division multiplexing (GFDM), fixed mobile convergence (FMC), universal fixed mobile convergence (UFMC), unique word OFDM (UW-OFDM), unique word DFT-spread OFDM (UW DFT-Spread-OFDM), cyclic prefix OFDM CP-OFDM, resource-block-filtered OFDM, Wi Fi, WLAN, WiMax, and the like. However, various features and functionalities of system **100** are particularly described wherein the devices of system **100** are configured to communicate wireless signals using one or more multi carrier modulation schemes, wherein data symbols can be transmitted simultaneously over multiple frequency subcarriers (e.g., OFDM, CP-OFDM, DFT-spread OFDM, UFMC, FMBC, etc.). The embodiments are applicable to single carrier as well as to multicarrier (MC) or carrier aggregation (CA) operation of the user equipment. The term carrier aggregation (CA) is also called (e.g., interchangeably called) “multi-carrier system”, “multi-cell operation”, “multi-carrier operation”, “multi-carrier” transmission and/or reception. Note that some embodiments are also applicable for Multi RAB (radio bearers) on some carriers (that is data plus speech is simultaneously scheduled).

Various embodiments described herein can be configured to provide and employ 5G wireless networking features and functionalities. With 5G networks that may use waveforms that split the bandwidth into several sub bands, different types of services can be accommodated in different sub bands with the most suitable waveform and numerology, leading to improved spectrum utilization for 5G networks. Notwithstanding, in the mmWave spectrum, the millimeter waves have shorter wavelengths relative to other communications waves, whereby mmWave signals can experience severe path loss, penetration loss, and fading. However, the shorter wavelength at mmWave frequencies also allows more antennas to be packed in the same physical dimension, which allows for large-scale spatial multiplexing and highly directional beamforming.

FIG. **10** provides additional context for various embodiments described herein, intended to provide a brief, general description of a suitable operating environment **1000** in which the various embodiments of the embodiment described herein can be implemented. While the embodiments have been described above in the general context of computer-executable instructions that can run on one or more computers, those skilled in the art will recognize that the embodiments can be also implemented in combination with other program modules and/or as a combination of hardware and software.

Generally, program modules include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the various methods can be practiced with other computer system configurations, including single-processor or multiprocessor computer systems, minicomputers, mainframe computers, Internet of Things (IoT) devices, distributed computing systems, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network.

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In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media or machine-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data or unstructured data.

Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

With reference again to FIG. **10**, the example operating environment **1000** for implementing various embodiments of the aspects described herein includes a computer **1002**, the computer **1002** including a processing unit **1004**, a system memory **1006** and a system bus **1008**. The system bus **1008** couples system components including, but not limited to, the system memory **1006** to the processing unit **1004**. The processing unit **1004** can be any of various commercially available processors. Dual microprocessors and other multi-processor architectures can also be employed as the processing unit **1004**.

The system bus **1008** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **1006** includes ROM

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**1010** and RAM **1012**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **1002**, such as during startup. The RAM **1012** can also include a high-speed RAM such as static RAM for caching data.

The computer **1002** further includes an internal hard disk drive (HDD) **1014** (e.g., EIDE, SATA), one or more external storage devices **1016** (e.g., a magnetic floppy disk drive (FDD) **1016**, a memory stick or flash drive reader, a memory card reader, etc.) and a drive **1020**, e.g., such as a solid-state drive, an optical disk drive, which can read or write from a disk **1022**, such as a CD-ROM disc, a DVD, a BD, etc. Alternatively, where a solid-state drive is involved, disk **1022** would not be included, unless separate. While the internal HDD **1014** is illustrated as located within the computer **1002**, the internal HDD **1014** can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment **1000**, a solid-state drive (SSD) could be used in addition to, or in place of, an HDD **1014**. The HDD **1014**, external storage device(s) **1016** and drive **1020** can be connected to the system bus **1008** by an HDD interface **1024**, an external storage interface **1026** and a drive interface **1028**, respectively. The interface **1024** for external drive implementations can include at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **1002**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

A number of program modules can be stored in the drives and RAM **1012**, including an operating system **1030**, one or more application programs **1032**, other program modules **1034** and program data **1036**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **1012**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

Computer **1002** can optionally comprise emulation technologies. For example, a hypervisor (not shown) or other intermediary can emulate a hardware environment for operating system **1030**, and the emulated hardware can optionally be different from the hardware illustrated in FIG. 10. In such an embodiment, operating system **1030** can comprise one virtual machine (VM) of multiple VMs hosted at computer **1002**. Furthermore, operating system **1030** can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications **1032**. Runtime environments are consistent execution environments that allow applications **1032** to run on any operating system that includes the runtime environment. Similarly, operating

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system **1030** can support containers, and applications **1032** can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

Further, computer **1002** can be enable with a security module, such as a trusted processing module (TPM). For instance, with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer **1002**, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

A user can enter commands and information into the computer **1002** through one or more wired/wireless input devices, e.g., a keyboard **1038**, a touch screen **1040**, and a pointing device, such as a mouse **1042**. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit **1004** through an input device interface **1044** that can be coupled to the system bus **1008**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a USB port, an IR interface, a BLUETOOTH® interface, etc.

A monitor **1046** or other type of display device can be also connected to the system bus **1008** via an interface, such as a video adapter **1048**. In addition to the monitor **1046**, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

The computer **1002** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **1050**. The remote computer(s) **1050** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **1002**, although, for purposes of brevity, only a memory/storage device **1052** is illustrated. The logical connections depicted include wired/wireless connectivity to a local area network (LAN) **1054** and/or larger networks, e.g., a wide area network (WAN) **1056**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

When used in a LAN networking environment, the computer **1002** can be connected to the local network **1054** through a wired and/or wireless communication network interface or adapter **1058**. The adapter **1058** can facilitate wired or wireless communication to the LAN **1054**, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter **1058** in a wireless mode.

When used in a WAN networking environment, the computer **1002** can include a modem **1060** or can be connected to a communications server on the WAN **1056** via other means for establishing communications over the WAN

1056, such as by way of the Internet. The modem 1060, which can be internal or external and a wired or wireless device, can be connected to the system bus 1008 via the input device interface 1044. In a networked environment, program modules depicted relative to the computer 1002 or portions thereof, can be stored in the remote memory/storage device 1052. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

When used in either a LAN or WAN networking environment, the computer 1002 can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices 1016 as described above, such as but not limited to a network virtual machine providing one or more aspects of storage or processing of information. Generally, a connection between the computer 1002 and a cloud storage system can be established over a LAN 1054 or WAN 1056 e.g., by the adapter 1058 or modem 1060, respectively. Upon connecting the computer 1002 to an associated cloud storage system, the external storage interface 1026 can, with the aid of the adapter 1058 and/or modem 1060, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface 1026 can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer 1002.

The computer 1002 can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

The above description of illustrated embodiments of the subject disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosed embodiments to the precise forms disclosed. While specific embodiments and examples are described herein for illustrative purposes, various modifications are possible that are considered within the scope of such embodiments and examples, as those skilled in the relevant art can recognize.

In this regard, while the disclosed subject matter has been described in connection with various embodiments and corresponding Figures, where applicable, it is to be understood that other similar embodiments can be used or modifications and additions can be made to the described embodiments for performing the same, similar, alternative, or substitute function of the disclosed subject matter without deviating therefrom. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims below.

Further to the description above, as it employed in the subject specification, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel

platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor may also be implemented as a combination of computing processing units.

In the subject specification, terms such as “store,” “storage,” “data store,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can include both volatile and nonvolatile memory.

As used in this application, the terms “component,” “system,” “platform,” “layer,” “selector,” “interface,” and the like are intended to refer to a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media, device readable storage devices, or machine-readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can include a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components.

In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. Moreover, articles “a” and “an” as used in the subject specification and annexed drawings

should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

Additionally, the terms “core-network”, “core”, “core carrier network”, “carrier-side”, or similar terms can refer to components of a telecommunications network that typically provides some or all of aggregation, authentication, call control and switching, charging, service invocation, or gateways. Aggregation can refer to the highest level of aggregation in a service provider network wherein the next level in the hierarchy under the core nodes is the distribution networks and then the edge networks. User equipment do not normally connect directly to the core networks of a large service provider, but can be routed to the core by way of a switch or radio area network. Authentication can refer to determinations regarding whether the user requesting a service from the telecom network is authorized to do so within this network or not. Call control and switching can refer to determinations related to the future course of a call stream across carrier equipment based on the call signal processing. Charging can be related to the collation and processing of charging data generated by various network nodes. Two common types of charging mechanisms found in present day networks can be prepaid charging and postpaid charging. Service invocation can occur based on some explicit action (e.g., call transfer) or implicitly (e.g., call waiting). It is to be noted that service “execution” may or may not be a core network functionality as third-party network/nodes may take part in actual service execution. A gateway can be present in the core network to access other networks. Gateway functionality can be dependent on the type of the interface with another network.

Furthermore, the terms “user,” “subscriber,” “customer,” “consumer,” “prosumer,” “agent,” and the like are employed interchangeably throughout the subject specification, unless context warrants particular distinction(s) among the terms. It should be appreciated that such terms can refer to human entities or automated components (e.g., supported through artificial intelligence, as through a capacity to make inferences based on complex mathematical formalisms), that can provide simulated vision, sound recognition and so forth.

Aspects, features, or advantages of the subject matter can be exploited in substantially any, or any, wired, broadcast, wireless telecommunication, radio technology or network, or combinations thereof. Non-limiting examples of such technologies or networks include Geocast technology; broadcast technologies (e.g., sub-Hz, ELF, VLF, LF, MF, HF, VHF, UHF, SHF, THz broadcasts, etc.); Ethernet; X.25; powerline-type networking (e.g., PowerLine AV Ethernet, etc.); femto-cell technology; Wi-Fi; Worldwide Interoperability for Microwave Access (WiMAX); Enhanced General Packet Radio Service (Enhanced GPRS); Third Generation Partnership Project (3GPP or 3G) Long Term Evolution (LTE); 3GPP Universal Mobile Telecommunications System (UMTS) or 3GPP UMTS; Third Generation Partnership Project 2 (3GPP2) Ultra Mobile Broadband (UMB); High Speed Packet Access (HSPA); High Speed Downlink Packet Access (HSDPA); High Speed Uplink Packet Access (HSUPA); GSM Enhanced Data Rates for GSM Evolution (EDGE) Radio Access Network (RAN) or GERAN; Terrestrial Radio Access Network (UTRAN); or LTE Advanced.

What has been described above includes examples of systems and methods illustrative of the disclosed subject matter. It is, of course, not possible to describe every combination of components or methods herein. One of ordinary skill in the art may recognize that many further combinations and permutations of the disclosure are pos-

sible. Furthermore, to the extent that the terms “includes,” “has,” “possesses,” and the like are used in the detailed description, claims, appendices and drawings such terms are intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

While the various embodiments are susceptible to various modifications and alternative constructions, certain illustrated implementations thereof are shown in the drawings and have been described above in detail. It should be understood, however, that there is no intention to limit the various embodiments to the specific forms disclosed, but on the contrary, the intention is to cover all modifications, alternative constructions, and equivalents falling within the spirit and scope of the various embodiments.

In addition to the various implementations described herein, it is to be understood that other similar implementations can be used, or modifications and additions can be made to the described implementation(s) for performing the same or equivalent function of the corresponding implementation(s) without deviating therefrom. Still further, multiple processing chips or multiple devices can share the performance of one or more functions described herein, and similarly, storage can be affected across a plurality of devices. Accordingly, the embodiments are not to be limited to any single implementation, but rather are to be construed in breadth, spirit and scope in accordance with the appended claims.

What is claimed is:

1. A method, comprising:

identifying, by source node equipment comprising a processor, a first route on a network from the source node equipment to destination node equipment via first node equipment and a second route on the network from the source node equipment to the destination node equipment via second node equipment;

receiving, by the source node equipment, from the first and the second node equipment respectively, a first predictive model comprising a first function that is used to predict a first delay predicted for the first route and a second predictive model comprising a second function that is used to predict a second delay predicted for the second route; and

based on the first function of the first predictive model and the second function of the second predictive model, selecting, by the source node equipment, a route from a group of routes, comprising the first route and the second route, for communication of information to the destination node equipment from the source node equipment, resulting in a selected route.

2. The method of claim 1, further comprising, based on the first predictive model and the second predictive model, generating, by the source node equipment, a third predictive model comprising a third function that is used to predict a third delay predicted for the communication of the information to the destination node equipment via the source node equipment.

3. The method of claim 2, wherein the selecting the route is further based on the third predictive model.

4. The method of claim 2, wherein the third predictive model comprises a neural network trained with data describing the network.

5. The method of claim 4, further comprising:

receiving, by the source node equipment, from node equipment implicated by communication using the

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selected route, route information corresponding to a result of the communication of the information via the selected route; and

updating, by the source node equipment, the third predictive model based on the route information.

6. The method of claim 5, wherein the updating the third predictive model comprises further training the neural network based on the route information.

7. The method of claim 1, wherein the first predictive model was generated by the first node equipment based on route information collected by the first node equipment.

8. The method of claim 7, wherein the first predictive model was generated by the first node equipment further based on the route information describing routes similar to the first route.

9. The method of claim 7, wherein the route information collected by the first node equipment comprises historical information describing past performance communicating via the first route.

10. The method of claim 1, wherein the first predictive model comprises information describing a third route, between the first node equipment and the destination node equipment.

11. A machine-readable storage medium, comprising executable instructions that, when executed by a processor of a first routing device, facilitate performance of operations, the operations comprising:

identifying a first routing network path from the first routing device to a destination device via a first network router and a second routing network path from the first routing device to the destination device via a second network router;

receiving, from the first network router and the second network router, respectively, a first predictive model comprising a first function that is used to predict a first delay predicted for the first routing network path and a second predictive model comprising a second function that is used to predict a second delay predicted for the second routing network path; and

based on the first function of the first predictive model and the second function of the second predictive model, selecting the first routing network path for communication of a transmission control protocol packet to the destination device from the first routing device, resulting in a selected network path.

12. The machine-readable storage medium of claim 11, wherein the operations further comprise, based on the first predictive model and the second predictive model, identifying a third predictive model comprising a third function that is used to predict a third delay predicted for communication of information to the destination device via the first routing device.

13. The machine-readable storage medium of claim 12, wherein the selecting the first routing network path is further based on the third predictive model.

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14. The machine-readable storage medium of claim 12, wherein the third predictive model comprises a model output from a neural network trained with data describing a network comprising the first routing network path and the second routing network path.

15. A source node equipment comprising:

a processor; and

a memory that stores executable instructions that, when executed by the processor, facilitate performance of operations, the operations comprising:

identifying a first route on a network from the source node equipment to destination node equipment via first node equipment and a second route on the network from the source node equipment to the destination node equipment via second node equipment;

receiving from the first and the second node equipment respectively, a first predictive model comprising a first function that is used to predict a first delay predicted for the first route and a second predictive model comprising a second function that is used to predict a second delay predicted for the second route; and

based on the first function of the first predictive model and the second function of the second predictive model, selecting a route from a group of routes comprising the first route and the second route, for communication of information to the destination node equipment from the source node equipment, resulting in a selected route.

16. The source node equipment of claim 15, the operations further comprising, based on the first predictive model and the second predictive model, generating a third predictive model comprising a third function that is used to predict a third delay predicted for the communication of the information to the destination node equipment via the source node equipment.

17. The source node equipment of claim 16, wherein the selecting the route is further based on the third predictive model.

18. The source node equipment of claim 16, wherein the third predictive model comprises a neural network trained with data describing the network.

19. The source node equipment of claim 18, the operations further comprising:

receiving from node equipment implicated by communication using the selected route, route information corresponding to a result of the communication of the information via the selected route; and

updating the third predictive model based on the route information.

20. The source node equipment of claim 19, wherein the updating the third predictive model comprises further training the neural network based on the route information.

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